Today

- More on properties of solutions of time-independent Schrödinger equation
- One-dimensional systems
- The infinite square well
- The simple harmonic oscillator

1.15.4 Spatial part of stationary wavefunction can be chosen to be real

Given $\psi_E(\vec{r})$ is a solution to the time-independent Schrodinger equation with energy E

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.6}$$

and complex conjugating it, we have

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r})^* + V(\vec{r})\psi_E(\vec{r})^* = E\psi_E(\vec{r})^*. \tag{1.15.7}$$

In other words, $\psi_E(\vec{r})^*$ is also a solution with the same energy. Therefore, we can trade $\psi_E(\vec{r})$ and $\psi_E(\vec{r})^*$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(\vec{r})^* \qquad \text{and} \qquad \frac{\psi_E(\vec{r}) - \psi_E(\vec{r})^*}{i}, \tag{1.15.8}$$

which are both real. Thus, the spatial part of a stationary wavefunction can be chosen to be real.

1.15.5 Parity symmetry: Even and odd wavefunctions

Suppose the potential is symmetric under reflection about the origin, $V(-\vec{r}) = V(\vec{r})$. In this case, given

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.9}$$

and replacing $\vec{r}$ by $-\vec{r}$ and setting $V(-\vec{r}) = V(\vec{r})$, we have a

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(-\vec{r}) + V(\vec{r})\psi_E(-\vec{r}) = E\psi_E(-\vec{r}). \tag{1.15.10}$$

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Therefore, $\psi_E(-\vec{r})$ is a solution with the same energy. We can trade $\psi_E(\vec{r})$ and $\psi_E(-\vec{r})$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(-\vec{r}) \qquad \text{and} \qquad \psi_E(\vec{r}) - \psi_E(-\vec{r}). \tag{1.15.11}$$

Now $\psi_E(\vec{r}) + \psi_E(-\vec{r})$ is even under reflection while $\psi_E(\vec{r}) - \psi_E(-\vec{r})$ is odd. Thus, when the potential is symmetric under reflection through the origin, we can choose the stationary states to be either even or odd under reflection.

1.15.6 Orthogonality/orthonormality

Suppose $\psi_n(\vec{r}, t)$ is a solution with energy E_n and $\psi_m(\vec{r}, t)$ is a solution with energy E_m; i.e.,

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_n(\vec{r}) = E_n\psi_n(\vec{r}) \quad \text{and} \quad -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_m(\vec{r})^* = E_m\psi_m(\vec{r})^*, \tag{1.15.12}$$

where we have taken the complex conjugate for the second. Multiplying both sides of the first by $\psi_m(\vec{r})^*$ and the second by $\psi_n(\vec{r})$,

$$\begin{aligned} -\frac{\hbar^2}{2m}\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_m(\vec{r})^*\psi_n(\vec{r}) &= E_n\psi_m(\vec{r})^*\psi_n(\vec{r}) \\ -\frac{\hbar^2}{2m}\psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_n(\vec{r})\psi_m(\vec{r})^* &= E_m\psi_n(\vec{r})\psi_m(\vec{r})^* \end{aligned} \tag{1.15.13}$$

Subtracting the two,

$$-\frac{\hbar^2}{2m}\left(\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) - \psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^*\right) = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}) \tag{1.15.14}$$

Recalling our computation and definition of the probability current, the left hand side equals $-i\hbar\vec{\nabla}\cdot\vec{J}$. Therefore,

$$-i\hbar\vec{\nabla}\cdot\vec{J} = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}). \tag{1.15.15}$$

Integrating both sides, and using the divergence theorem, $\int \vec{\nabla}\cdot\vec{J}d^3\vec{r} = \oint \vec{J}\cdot d\vec{a}$,

$$-i\hbar\oint \vec{J}\cdot d\vec{a} = (E_n - E_m)\int \psi_m(\vec{r})^*\psi_n(\vec{r})d^3\vec{r}. \tag{1.15.16}$$

Now, taking the boundary of the area integral to infinity, the current vanishes, since a normalizable wavefunction and the corresponding current must vanish at infinity, and the left hand

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side vanishes. We then have

$$(E_n - E_m) \int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 0. \tag{1.15.17}$$

We have two cases.

1. Case 1: $n = m$ and $E_n = E_m$. In this case, $E_n - E_m = 0$ and for a normalized wavefunction $\psi_n(\vec{r})$, $\int \psi_n(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 1$.
2. Case 2: $n \neq m$ and $E_n \neq E_m$. In this case, $E_n - E_m \neq 0$ and $\int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = 0$. We say $\psi_n(\vec{r})$ and $\psi_m(\vec{r})$ are orthogonal.

Combining the two cases, we have the orthonormality relation

$$\int \psi_m(\vec{r})^* \psi_n(\vec{r}) d^3\vec{r} = \delta_{mn}, \tag{1.15.18}$$

where

$$\delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases}$$

is called the Kronecker delta.

1.15.7 Linearity

The Schrödinger equation is linear. Given stationary states $\psi_n(\vec{r}, t) = \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, $n = 1, 2, ...$, a linear combination of these

$$\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}, \tag{1.15.19}$$

where c_n are complex constants, is a solution to the time-dependent Schrödinger equation $i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = \hat{H}\psi(\vec{r}, t)$. To check that, the left hand side is $i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = \sum_n c_n \psi_n(\vec{r}) i\hbar \frac{\partial}{\partial t} e^{-\frac{i}{\hbar} E_n t} = \sum_n c_n E_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, and the right hand side is $\hat{H}\psi(\vec{r}, t) = \sum_n c_n \hat{H}\psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t} = \sum_n c_n E_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar} E_n t}$, and they are equal.

1.15.8 Time evolution

Question 1.15.2: Given stationary states ψ_1 and ψ_2 with different energies, is $c_1\psi_1 + c\psi_2$ stationary? (A). Yes, (B). No.

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Because $\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}$ with constant coefficients c_n is a solution to the time-dependent Schrödinger equation, and $\psi(\vec{r},0) = \sum_n c_n\psi_n(\vec{r},0) = \sum_n c_n\psi_n(\vec{r})$ at time $t = 0$, we note that given

$$\psi(\vec{r},0) = \sum_n c_n\psi_n(\vec{r},0) = \sum_n c_n\psi_n(\vec{r}) \tag{1.15.20}$$

at time $t = 0$, the time evolution, or the wavefunction at time t, is simply

$$\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}. \tag{1.15.21}$$

In other words, once we have expanded a given initial wavefunction in terms of a linear combination of the stationary wavefunctions $\psi_n(\vec{r})$, the time evolution follows simply by putting a factor of $e^{-\frac{i}{\hbar}E_nt}$ to each term containing $\psi_n(\vec{r})$.

1.15.9 Normalization

Given $\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}$ with complex coefficients c_n, normalizing it leads to

$$\begin{aligned} 1 &= \int |\psi(\vec{r},t)|^2 d^3\vec{r} = \int \left(\sum_m c_m^*\psi_m^*(\vec{r})e^{\frac{i}{\hbar}E_mt}\right)\left(\sum_n \psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}\right) d^3\vec{r} \\ &= \sum_n\sum_m c_m^*c_n e^{\frac{i}{\hbar}(E_m-E_n)t}\int \psi_m^*(\vec{r})\psi_n(\vec{r})d^3\vec{r} = \sum_n\sum_m c_m^*c_n e^{\frac{i}{\hbar}(E_m-E_n)t}\delta_{mn} = \sum_n |c_n|^2. \end{aligned} \tag{1.15.22}$$

Therefore, the constant coefficients are constrained by

$$\sum_n |c_n|^2 = 1. \tag{1.15.23}$$

1.15.10 Completeness

Given an arbitrary $\psi(\vec{r},t)$ that is a solution to the Schrödinger equation with certain boundary conditions, we can express it as a linear superposition of the stationary solutions $\psi_n(\vec{r},t) = \psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}$ as

$$\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_nt}. \tag{1.15.24}$$

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with complex coefficients c_n. In other words, the stationary orthonormal solutions form a complete set.

To find the condition for the above statement to hold, first let us find the coefficients c_n. Multiplying both sides by $\psi_m(\vec{r},t)^*$ and integrating over $d^3\vec{r}$,

$$\int \psi_m(\vec{r},t)^*\psi(\vec{r},t)d^3\vec{r} = \sum_n c_n \int \psi_m(\vec{r},t)^*\psi_n(\vec{r},t)d^3\vec{r} = \sum_n c_n\delta_{mn} = c_m. \tag{1.15.25}$$

Therefore,

$$c_n = \int \psi_n(\vec{r},t)^*\psi(\vec{r},t)d^3\vec{r}. \tag{1.15.26}$$

Inserting these coefficients back into the expansion,

$$\begin{aligned}\psi(\vec{r},t) &= \sum_n \int \psi_n(\vec{r}',t)^*\psi(\vec{r}',t)d^3\vec{r}'\psi_n(\vec{r},t) \\ &= \int d^3\vec{r}'\psi(\vec{r}',t)\sum_n \psi_n(\vec{r}',t)^*\psi_n(\vec{r},t).\end{aligned} \tag{1.15.27}$$

The stationary states form a complete set if

$$\sum_n \psi_n(\vec{r}',t)^*\psi_n(\vec{r},t) = \delta^3(\vec{r}'-\vec{r}), \tag{1.15.28}$$

where $\delta^3(\vec{r}'-\vec{r})$ is the Dirac-delta function in three dimensions defined by

$$\int d^3\vec{r}'\psi(\vec{r}',t)\delta^3(\vec{r}'-\vec{r}) = \psi(\vec{r},t). \tag{1.15.29}$$

In other words, if we multiply and sum the stationary states as $\sum_n \psi_n(\vec{r}',t)^*\psi_n(\vec{r},t)$ and we get $\delta^3(\vec{r}'-\vec{r})$, we know we have a complete set. Conversely, if we miss a state in the sum, we do not get $\delta^3(\vec{r}'-\vec{r})$. In cartesian coordinates, $\delta^3(\vec{r}'-\vec{r}) = \delta(x-x')\delta(y-y')\delta(z-z')$.

1.16 Delta function

Below, we summarize some of the properties of the delta function, focusing on one dimension. Given a function $f(x)$, the delta function $\delta(x-x')$ is defined such that

$$f(x') = \int f(x)\delta(x-x')dx \tag{1.16.1}$$

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In other words, $\delta(x-x')$ in the integral projects out the value of $f(x)$ at the zero of the argument $x-x'$ of $\delta(x-x')$. Therefore, we can view $\delta(x-x')$ as defined by

$$\delta(x-x')=0 \text{ for } x\neq x', \qquad f(x')=\int_a^b f(x)\delta(x-x')dx \text{ for } a<x' \text{ and } b>x'. \tag{1.16.2}$$

Because $\delta(x-x')=0$ for $x\neq x'$, we can rewrite $f(x')=\int f(x)\delta(x-x')dx=f(x')\int\delta(x-x')dx$. Therefore,

$$\int \delta(x-x')dx=1. \tag{1.16.3}$$

However, the delta function is not actually a function but a distribution. For instance, we can not interpret $\int\delta(x-x')dx=1$ as the area under $\delta(x-x')$ being one.

Here we list some representations of the delta function,

$$\delta_\alpha(x) = \frac{1}{\alpha\sqrt{\pi}}e^{-\frac{x^2}{a^2}} \tag{1.16.4}$$

$$\delta_\alpha(x) = \frac{1}{\pi x}\sin\left(\frac{x}{\alpha}\right) \tag{1.16.5}$$

$$\delta_\alpha(x) = \frac{\alpha}{\pi x^2}\sin^2\left(\frac{x}{\alpha}\right) \tag{1.16.6}$$

With any of these representations, the delta function $\delta_\alpha(x-a)$ is defined through

$$f(x')=\lim_{\alpha\to 0}\int f(x)\delta_\alpha(x-x')dx. \tag{1.16.7}$$

However, $\lim_{\alpha\to0}\int f(x)\delta_\alpha(x-x')dx\neq\int\lim_{\alpha\to0}f(x)\delta_\alpha(x-x')dx$. Indeed the integral vanishes if the limit is taken within the integral.

Exercise 1.16.1: Show the following.

1. $\delta(ax)=\frac{1}{|a|}\delta(x)$
2. $\delta(x^2-a^2)=\frac{1}{2|a|}\left(\delta(x+a)+\delta(x-a)\right)$
3. $\delta(f(x))=\sum_i\frac{\delta(x-x_i)}{|f'(x)|}$, where x_i are the roots of $f(x)=0$.
4. Write $\delta^3(\vec{r}-\vec{r}')$ in spherical coordinates.
5. Write $\delta^3(\vec{r}-\vec{r}')$ in cylindrical coordinates.

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2 One-dimensional systems

Now, we begin constructing the wavefunctons for a system containing a single particle of mass m in one dimension. Different systems correspond to different potentials. We will be considering only potentials that are time-independent. The wavefunctions and the corresponding energies are then obtained by solving the time-independent Schrödinger equation in one dimension,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x). \tag{2.0.1}$$

Once we find the wavefunction $\psi_E(x)$ of energy E, its time dependence follows easily,

$$\psi_E(x,t) = \psi_E(x)e^{-\frac{i}{\hbar}Et}. \tag{2.0.2}$$

2.1 Boundary conditions

The boundary conditions we impose depend on the potential. Suppose we are given a generic potential $V(x)$. Consider a point $x = a$ on the $x-$axis. We want to investigate whether the wavefunction $\psi(x)$ and its first derivative $\psi'(x)$ are continuous or not across $x = a$. First, rewriting the Schrödnger equation as

$$\psi''(x) = \frac{2m}{\hbar^2}(V(x) - E)\psi(x), \tag{2.1.1}$$

we note that the continuity or discontinuity of $\psi''(x)$ follows the continuity or discontinuity of the potential $V(x)$.

For the first derivative, and considering a particle of finite energy E,

$$\begin{aligned}\lim_{\epsilon\to 0}(\psi'(a+\epsilon) - \psi'(a-\epsilon)) &= \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\frac{2m}{\hbar^2}(V(x)-E)\psi(x)dx \\ &= \lim_{\epsilon\to 0}\frac{2m}{\hbar^2}\int_{a-\epsilon}^{a+\epsilon}V(x)\psi(x)dx,\end{aligned} \tag{2.1.2}$$

Now if $V(x)$ has a finite jump at $x = a$, this integral vanishes and $\psi'(x)$ is continuous across $x = a$. On the other hand, if $V(x)$ has an infinite jump, similar to the delta function, the integral will be nonzero but finite, and $\psi'(x)$ is discontinuous across $x = a$.

For the wavefunction itself,

$$\lim_{\epsilon\to 0}(\psi(a+\epsilon) - \psi(a-\epsilon)) = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi'(x)dx. \tag{2.1.3}$$

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Now whether $V(x)$ has a finite or infinite jump at $x = a$, this integral vanishes, since $\psi'(x)$ could have only a finite jump across $x = a$. Therefore, $\psi(x)$ is always continuous across $x = a$.

We summarize the boundary conditions as follows

1. When the potential $V(x)$ has a finite jump at $x = a$, both $\psi(x)$ and $\psi'(x)$ are continuous across $x = a$.
2. When the potential $V(x)$ has an infinite jump at $x = a$, $\psi(x)$ is continuous but $\psi'(x)$ is discontinuous across $x = a$.

Furthermore, the wavefunction must vanish at $x = \pm\infty$ for a normalizable wavefunction.

2.2 Infinite square well

Consider a particle of mass m in the potential

$$V(x) = \begin{cases} 0, & 0 \le x \le a \\ \infty, & x < 0 \text{ and } x > a. \end{cases}$$

The potential is shown in the figure below.

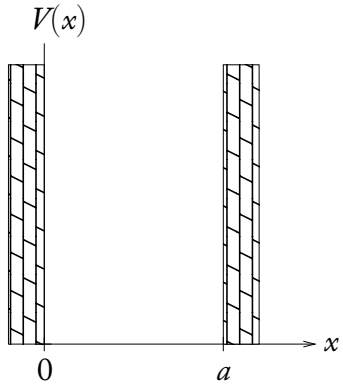


Figure 2.2.1: Infinite square well potential.

Because, the particle cannot have infinite energy, it cannot be found in the regions $x < 0$ and $x > a$. Therefore,

$$\psi(x) = 0, \quad \text{for } x < 0 \text{and } x > a. \tag{2.2.1}$$

For the region in the well, $0 \le x \le a$, we have $V(x) = 0$ and the Schrödinger equation reduces

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to

$$\psi''(x) + k^2\psi(x) = 0, \tag{2.2.2}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \tag{2.2.3}$$

and $E > 0$. We can write the general solution in terms of linear combinations of real solutions $\sin kx$ and $\cos kx$ as

$$\psi(x) = A\sin kx + B\cos kx. \tag{2.2.4}$$

Applying the boundary condition on the continuity of $\psi(x)$ at $x = 0$ sets $\psi(0) = B = 0$. We then have $\psi(x) = A\sin kx$, and applying the boundary condition at $x = a$ sets $\psi(a) = A\sin ka = 0$. Therefore, either $A = 0$ or $ka = n\pi$, $n = 1, 2, ...$ But $A = 0$ sets $\psi(x) = 0$ everywhere, which corresponds to no particle anywhere. Therefore,

$$k_n = \frac{n\pi}{a}, \quad n = 1, 2, ... \tag{2.2.5}$$

The wave vector is discrete, which means that both the momentum and the energy of the particle are discrete too, since $p = \hbar k$ and $E = \sqrt{\frac{2mE}{\hbar^2}}$. Therefore, $\psi_n(x) = A_n \sin k_n x = A_n \sin(\frac{n\pi}{a}x)$. Normalizing and choosing the wavefunctions to be real, $1 = \int_0^a |\psi_n(x)|^2 dx = \int_0^a A_n^2 \sin^2(\frac{n\pi}{a}x)dx = A_n^2 \frac{a}{2}$, or $A_n = \sqrt{\frac{2}{a}}$. The corresponding energies are $E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2\pi^2 n^2}{2ma^2}$. Summarizing, the normalized wavefunctions and the corresponding energies for the particle of mass m in the given infinite square well are

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin(\frac{n\pi}{a}x), \quad E_n = \frac{\hbar^2\pi^2 n^2}{2ma^2}, \qquad n = 1, 2, 3, ... \tag{2.2.6}$$

We ignore $n = 0$, since $\psi(x)$ is identically zero in this case. We ignore $n = -1, -2, ...$, since $\psi_{-n}(x) = -\psi_n(x)$, the two differ only by an overall sign and are physically identical with the same probability density. The wavefunctions and the corresponding energies are shown in the schematic figure below.

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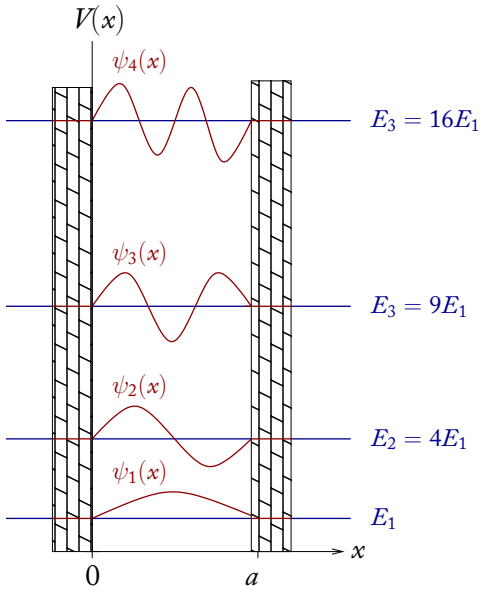


Figure 2.2.2: Wavefunctions and energies for a particle in the infinite square well potential.

Exercise 2.2.1:

1. Demonstrate that the particle cannot have $E = 0$ or $E < 0$ for the given potential well above the $x-$axis.
2. Show that $\psi(x) = 0$ outside the well by considering a finite potential of height V_0 and taking the limit $V_0 \to \infty$.

Solution:

1. If $E = 0$, then we have $\psi''(x) = 0$. This has the general solution $\psi(x) = Cx + D$. This with the boundary conditions $\psi(0) = \psi(a) = 0$ sets a straight line that passes through both $x = 0$ and $x = a$, or $\psi(x) = 0$, and there is no particle with $E = 0$. For $E < 0$, the Schrödinger equation becomes $\psi''(x) - \kappa^2\psi(x) = 0$, where $\kappa = \sqrt{-\frac{2mE}{\hbar^2}}$. The general solution is $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$. Applying the boundary conditions $\psi(0) = C + D = 0$ and $\psi(a) = Ce^{-\kappa a} + De^{\kappa a}$. Solving these, we get either $A = B = 0$ or $\kappa a = 0$. Therefore, we cannot have a particle with $E < 0$.
2. Considering a finite V_0, the general solution is $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$, where $\kappa = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$ now. Normalizability requires finiteness of the wavefunction at $x = \pm\infty$. Therefore, $\psi(x) = Ce^{-\kappa x}$ for $x > a$ and $\psi(x) = De^{\kappa x}$ for $x < 0$. Both vanish in the limit $V_0 \to \infty$ or $\kappa \to \infty$.

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Exercise 2.2.2:

1. The wavefunctions and energies for a particle of mass m in a one-dimensional infinite square well with potential $V(x) = 0$ within $0 \leq x \leq a$ and $V(x) = \infty$ outside are given by $\psi_n(x) = \sqrt{\frac{2}{a}} \sin(\frac{n\pi}{a}x)$ for $0 \leq x \leq a$, $\psi_n(x) = 0$ outside, and $E_n = \frac{\hbar^2\pi^2n^2}{2ma^2}$, where $n = 1, 2, ...$

   (a) Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and Δx in state $\psi_n(x)$. Does Δx increase, decrease, or stay the same with increasing n?

   (b) Determine $\langle\hat{p}\rangle$, $\langle\hat{p}^2\rangle$, and Δp in state $\psi_n(x)$. Does Δp increase, decrease, or stay the same with increasing n?

   (c) Check whether the uncertainty principle holds for $\Delta x\Delta p$ in each state $\psi_n(x)$. Does $\Delta x\Delta p$ increase, decrease, or stay the same with increasing n? (5 points)

   (d) What is ΔE in state $\psi_n(x)$?

   (e) Determine $[\hat{x}, \hat{H}]$. Point out whether $\psi(x) = \hat{x}\psi_n(x)$ is a stationary state or not without explicitly determining $\psi(x, t)$.

   (f) The particle is in state $\psi(x, 0) = \psi_n(x)$ at time $t = 0$.

      i. What is the probability that the particle is found in the left half of the well within $0 \leq x \leq \frac{a}{2}$ at time $t = 0$?

      ii. What is the probability that the particle is found in the left quadrant of the well within $0 \leq x \leq \frac{a}{4}$ at time $t = 0$? Interpret and discuss the difference between even n and odd n.

      iii. What is the state $\psi(x, t)$ and the average energy of the particle at later time t? Do your answers for (1(f)i) and (1(f)ii) change at time t?

   (g) The particle is in state $\psi(x, 0) = c_1\psi_1(x) + c_2\psi_2(x)$ at time $t = 0$, where c_1 and c_2 are real constants.

      i. Write down the normalized $\psi(x, 0)$ and $\psi(x, t)$ at time t.

      ii. List all possible values of energy that a single measurement could produce at time t and the corresponding probabilities. What is the

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expectation value of the energy (or average energy) of the particle at time t?

iii. Show that the probability density for finding the particle at point x within the well at time t oscillates in time. Determine the frequency and the amplitude of oscillation, and point out how the frequency is related to the energies E_1 and E_2.

iv. Determine $\langle \hat{x} \rangle$ at time t.

v. Determine $\langle \hat{p} \rangle$ at time t.

(h) The particle is in state $\psi(x,0) = \sqrt{\frac{30}{a^5}}\, x(a-x)$ for $0 \le x \le a$ and $\psi(x,0) = 0$ outside the well at time $t = 0$.

i. Make a schematic plot of $\psi(x,0)$. Determine the state $\psi(x,t)$ of the particle at later time t.

ii. What are the possible values that a single measurement of energy would produce at time t and the corresponding probabilities. Point out the energy with the highest probability and interpret and discuss the reason or what it means.

(i) Rewrite the wavefunctions when the location of the well where $V(x) = 0$ is moved to within $-\frac{a}{2} \le x \le \frac{a}{2}$.

Question 2.2.1: For the wavefunctions of the infinite square well, (A). $\psi(x)$ is continuous, (B). $\psi'(x)$ is continuous, (C). $\psi''(x)$ is continuous.

Question 2.2.2: For a single particle in a symmetric potential, the ground state wavefunction is (A). symmetric. (B). antisymmetric. (C). either symmetric or antisymmetric.

Question 2.2.3: How many nodes does $\psi_n(x)$ of the infinite square well have in the interior $0 < x < a$? (A). 0. (B). 2. (C). n. (D). $n-1$. (E). $n+1$. (F). It cannot be determined.

Question 2.2.4: Given stationary states $\psi_1(x,t)$ and $\psi_2(x,t)$, is $\psi_1(x,t) + \psi_2(x,t)$ stationary? (A). Yes. (B). No.

Question 2.2.5: Given two solutions $\psi_1(x,t)$ and $\psi_2(x,t)$ to the time-dependent Schrödinger equation, $\psi_1(x,t)+\psi_2(x,t)$ is (A). a solution to only the time-independent

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Schrödinger equation. (B). a solution to only the time-dependent Schrödinger equation. (C). a solution to both the time-independent and the time-dependent Schrödinger equation. (D). a solution to neither the time-independent nor the time-dependent Schrödinger equation.

Question 2.2.6: Given $\psi_1(x,0) = \hat{x}\psi_n(x)$ for the infinite square well, (A). $\psi(x,t) = x\psi_n(x)e^{-i\frac{E_n}{\hbar}t}$. (B). $\psi(x,t) = x\psi_n(x)e^{i\frac{E_n}{\hbar}t}$. (C). $\psi(x,t) = x\psi_n(x)$. (D). None of the above.

Question 2.2.7: Assuming that a quantum dot can be approximately described by an electron in a one-dimensional infinite square well potential, which solution of quantum dots has a well of largest width?



Figure 2.2.3: Quantum dots

(A). Red. (B). Orange. (C). Green. (D). Blue. (E). Violet.

2.3 The simple harmonic oscillator

Now we want to investigate a particle of mass m in the simple harmonic potential

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2. \tag{2.3.1}$$

Because the potential is time-independent, we start with solving the time-independent Schrödinger equation,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x) = E\psi(x). \tag{2.3.2}$$

We still need boundary conditions. Because the potential becomes infinite at $x = \pm\infty$, the particle cannot be found there, since it would need infinite energy. Alternatively, because the wavefunction needs to be normalizable, it has to vanish at $x = \pm\infty$. Therefore the boundary

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conditions are

$$\psi(-\infty) = 0, \qquad \psi(+\infty) = 0. \tag{2.3.3}$$

The wavefunction starts with $\psi(-\infty) = 0$ at $x = -\infty$, oscillates according to the Schrödinger equation in the interior, and falls back to $\psi(+\infty) = 0$ at $x = +\infty$. This equation is not easy to solve immediately. Therefore, we follow a number of steps.

Step 1: Simplify notation with change of variable

The first step is to rewrite the equation in simpler form without too many variables floating around. To do that, we make a change of variable of the coordinate x and define

$$\xi = \sqrt{\frac{m\omega}{\hbar}}x. \tag{2.3.4}$$

We then have $\frac{d}{dx} = \frac{d\xi}{dx}\frac{d}{d\xi} = \sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}$ and $\frac{d^2}{dx^2} = \frac{m\omega}{\hbar}\frac{d^2}{d\xi^2}$. The Schrödinger equation then takes the form

$$\frac{d^2\psi(\xi)}{d\xi^2} = (\xi^2 - K)\psi(\xi), \tag{2.3.5}$$

where

$$K \equiv \frac{2E}{\hbar\omega}. \tag{2.3.6}$$

Step 2: Asymptotic behavior

If we have an approximate solution in some region of ξ, then we can write a solution ansatz that is a product of the approximate solution and an unknown function $H(\xi)$ that would allow to solve the general equation exactly. Hopefully, the equation for $H(\xi)$ is easier to solve. In particular, (2.3.5) is difficult because of the ξ^2 term. If we could have approximate solution that solves it at this order, then this term would be eliminated. Because the ξ^2 term dominates over the K term for large ξ, let us work in the large $\xi^2 >> K$ region, where the equation reduces to

$$\frac{d^2\psi(\xi)}{d\xi^2} \sim \xi^2\psi(\xi). \tag{2.3.7}$$

Now we can immediately write approximate solutions,

$$\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}, \qquad \psi(\xi) \sim e^{\frac{1}{2}\xi^2}. \tag{2.3.8}$$

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Because a solution that goes like $e^{\frac{1}{2}\xi^2}$ at large ξ cannot be normalizable, we discard it and keep only $\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}$. Note that this gives $\psi''(\xi) \sim \xi^2 e^{-\frac{1}{2}\xi^2} - e^{-\frac{1}{2}\xi^2}$ which approximately equals $\xi^2 e^{-\frac{1}{2}\xi^2}$, since it dominates over $-e^{-\frac{1}{2}\xi^2}$ for large ξ, and thus the approximate equation is approximately solved. That is all well, since we are not seeking exact solutions at this point. We then write an ansatz solution,

$$\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}; \tag{2.3.9}$$

we want to solve for $H(\xi)$ exactly and produce the exact $\psi(\xi)$. Differentiating this,

$$\frac{d^2\psi(\xi)}{d\xi^2} = H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.3.10}$$

Putting (2.3.9) and (2.3.10) in (2.3.5), we have

$$H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} = \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} - KH(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.3.11}$$

Observe that the ξ^2 terms cancel out, since the approximate solution $e^{-\frac{1}{2}\xi^2}$ solves the Schrödinger equation at leading order in large ξ.

Thus, the Schrödinger equation reduces to an equation for $H(\xi)$.

$$H''(\xi) - 2\xi H'(\xi) + (K-1)H(\xi) = 0. \tag{2.3.12}$$

This equation is called the Hermite equation.

Step 3: Method of power series

It is known in the theory of differential equations that if the equation is written with the coefficient of the highest order derivative set to one as in here, then the equation could be singular only at the singular points of the coefficients of the lower order derivative terms. The coefficients we have in (2.3.12) have no singularity and, therefore, $H(\xi)$ can have no singularities at finite ξ and a power series solution will have an infinite radius of convergence. Thus, we can write a power series solution of the form

$$H(\xi) = \sum_{n=0}^{\infty} a_n \xi^n. \tag{2.3.13}$$

The idea is then to rewrite (2.3.12) as $\sum[\ldots]_n\xi^n = 0$, with constant coefficients $[\ldots]_n$, which holds for all ξ only if each constant coefficient vanishes; i.e., $[\ldots]_n = 0$ for each n. Therefore, we seek

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to write each term in (2.3.12) as $\sum[\ldots]_n\xi^n = 0$ so that we can easily combine the terms together. First, $H'(\xi) = \sum_{n=0}^{\infty} na_n\xi^{n-1}$. Because what we have in (2.3.12) is $-2\xi H'(\xi) = -2\sum_{n=0}^{\infty} na_n\xi^n$, which has the desired form, we keep it as is. Next, taking the second derivative, $H''(\xi) = \sum_{n=0}^{\infty} n(n-1)a_n\xi^{n-2}$. Now this has ξ^{n-2} while we seek ξ^n so that we can combine the coefficients with the other terms. But the first two terms for $n = 0$ and $n = 1$ are zero, and we can start the sum with $n = 2$, $H''(\xi) = \sum_{n=2}^{\infty} n(n-1)a_n\xi^{n-2}$. Next we shift $n \to n+2$ and the sum starts at $n = 0$ as in the others, which we do by rewriting it as $H''(\xi) = \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}\xi^n$, and we have expanded it in terms of ξ^n too. Putting these in (2.3.12), we have

$$\sum \left[(n+2)(n+1)a_{n+2} - (2n+1-K)a_n\right]\xi^n = 0. \tag{2.3.14}$$

This holds for all ξ only if each and every coefficient of ξ^n is identically zero, which gives the recursion relation

$$a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)}a_n \tag{2.3.15}$$

Observe that there are only two independent coefficients, since starting with a_0, we recursively produce a_2, a_4, etc, which produces even solutions. If we start with a_1, we recursively obtain a_3, a_5, etc, which produces odd solutions.

Step 4: Infinite series produces a diverging function

For large n, we have

$$\frac{a_{n+2}}{a_n} \to \frac{2}{n} \tag{2.3.16}$$

But this is similar to the behavior in the series expansion of $\xi^m e^{\xi^2}$ for any m, which goes like e^{ξ^2} for large ξ. In other words, if we keep all infinite series of terms, then we would have $H(\xi) \sim e^{\xi^2}$ and $\psi(\xi) = H(\xi) \sim e^{-\frac{1}{2}\xi^2} \sim e^{\frac{1}{2}\xi^2}$, and the wavefunction would blow up at large ξ and would not be normalizable.

Step 5: Truncate series

We need to truncate the sum so that $H(\xi)$ would at most be a polynomial of some finite degree n, and then $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$ would be finite, since $e^{-\frac{1}{2}\xi^2}$ decays much stronger than any polynomial of finite degree at large ξ. We truncate the series at n by setting $a_{n+2} = 0$ while $a_n \neq 0$, which is achieved when $2n + 1 - K = 0$, or using the definition of K,

$$E_n = (n + \frac{1}{2})\hbar\omega, \tag{2.3.17}$$

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where we could choose to truncate the series at $n = 0$, $n = 1$, $n = 2$, etc, which produce polynomials of the same degree. These energies are the energy levels of the simple harmonic oscillator, and the corresponding polynomials are the Hermite polynomials (up to irrelevant overall constant factors).

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